Project: Causal machine learning

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Question 1 (Amber and Feng)

As the choice of employment is made based on income, not on eligibility, but the job/employer decides whether you are eligible to participate in a 401(k) plan, there is an arrow from income (I) to eligibility (E). There is also an arrow from income to net financial assets (Y) as the income is likely to determine the accumulation of these assets. Income is therefore considered to be a common cause of E and Y. The unmeasured preference (U) for saving affects both participation in a 401(k) plan (P) and the accumulation of assets (Y). C denotes covariates like age, family size, years of education and other possible variables listed in Question 2. These covariates are likely to guide decisions in deciding on a job offer based on income (e.g. it might depend on the size of the family for instance which salary you might prefer or need); therefore the arrow from C to I. The unmeasured preference for saving might also be affected by these covariates, as well as participation in a 401(k) plan and the accumulation of assets. Therefore also arrows from C to U, P and Y^e.
As we are focusing on the causal effect of eligibility (E) on the accumulation of assets (Y), we make a SWIG, where we intervene only in E. This will give us only 1 open path from E to Y^e, which is through income. By conditioning on income we close this path, resulting in the condition that $E \perp\!\!\!\perp Y^e | I$, which is exactly the conditional exchangeability assumption.

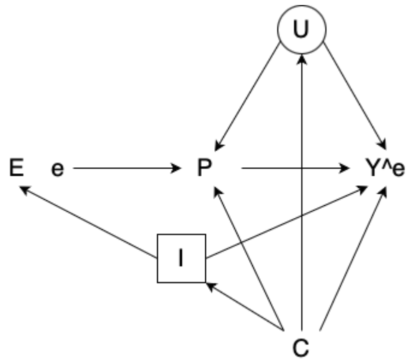


Figure 1: Causal diagram

Remark 1. *This remark may not be related to the solution to the first question but will give some insight for the following questions. Note that participation (P) is a collider of the paths E-P-U-Y, E-P-C-Y and E-P-C-U-Y. Therefore, we don't have to adjust for P when we consider the effect of eligibility (E) on the outcome (Y). Besides, when we condition on income (I), we also block the path E-I-C-Y, which implies eligibility for a 401(k) could be taken as exchangeable only conditional on income. This also means we don't have to condition on C as there are no more open paths from E to Y through C; conditioning on income is sufficient.*

Question 2 (Amber)

To obtain debiased (AIPW) estimators for the ATE and ATT, correcting for the desired covariates, we simply used the functions `ate` and `att` from the `npcausal` package, where we can perform 5-fold cross-fitting by setting the value of `nsplits` equal to 5. We explored 3 different algorithms from the SuperLearner library: gam, randomForest and glm.

To obtain estimates with TMLE, we manually created 5 folds to obtain initial estimates for

$\hat{E}[Y_i|A_i = 1, L_i]$ and $\hat{E}[Y_i|A_i = 0, L_i]$, and of the propensity score $\hat{P}(A_i = 1|L_i)$, using cross-fitting. For each fold, we trained the `SuperLearner` model on the remaining folds and made predictions for this held-out fold. These initial estimates were given as arguments in the function `tmle`. The resulting estimates and 95% confidence intervals for both AIPW and TMLE are shown in Table 1. We see in both approaches that we get highly positive effects. Eligibility for enrolling in a 401(k) plan increases the net financial assets with 7,930.24$ (AIPW) or 8,396.16$ (TMLE).

AIPW estimates

Effect	Estimate	CI (95%)
ATE	7,930.24	[5,783.65; 10,076.83]
ATT	10,705.09	[8,182.28; 13,227.90]

TMLE estimates

Effect	Estimate	CI (95%)
ATE	8 396.16	[6 218.74; 10 573.59]
ATT	11 354.72	[8 048.86; 14 660.59]

Table 1: AIPW estimates *(left)* and TMLE estimates *(right)* using 5-fold cross-fitting.

We notice a difference of about \$3,000 between ATE and ATT. However, looking at the ATT is more interesting when eligible and non-eligible populations are not comparable. A way to investigate this, is by exploring the propensity scores in the eligible and non-eligible individuals (see Figure 2). The propensity scores among the eligible observations, are more equally spread around 50%, where the majority of propensity scores for the non-eligible observations have a propensity score below 50%. There are no extremely small propensity scores (range from 0.05 to 0.86).

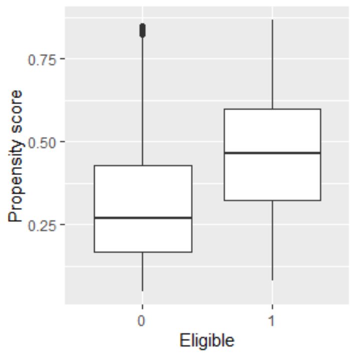


Figure 2: Propensity scores for the different groups of eligibility (TMLE)

Lastly, we can compare these estimates with the unadjusted estimate. For this purpose, we simply fit a linear regression model, only including the eligibility factor. This results in an effect size of 19,559.34$ ([16,999.90; 22,118.79]). This is clearly much larger than the ATE estimates when controlling the covariates.

Question 3 (Feng)

To test if eligibility for enrolling in a 401(k) plan can affect net financial assets, a more powerful test is proposed. Based on its efficient influence curve, we know that a debiased estimator of θ

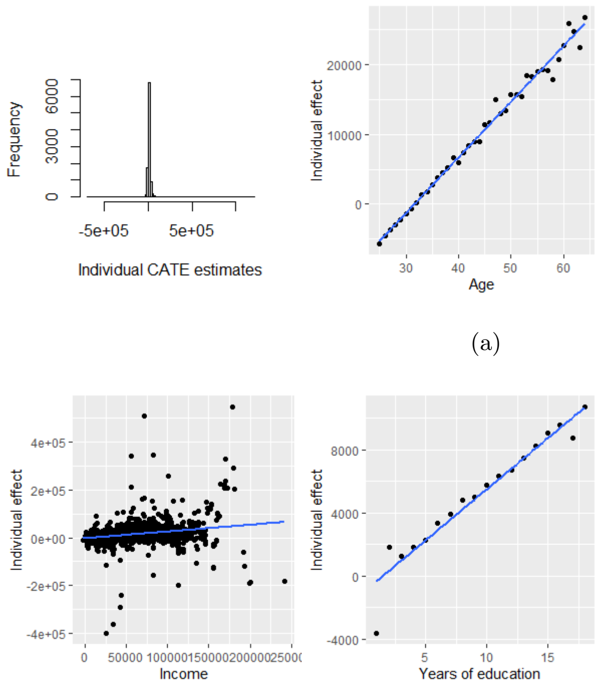
is

$$\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n}\{A_i - \hat{E}(A_i \mid L_i)\}\{Y_i - \hat{E}(Y_i \mid L_i)\},$$

where a 5-fold cross-fitting technique is used to obtain $\hat{\theta}$ in practice. The estimate of θ is 1816.52 and its standard error is 250.4039. Since the p-value of constructed t-test statistic is 2.018385e-13, we can reject the null hypothesis that θ equals 0. This claim aligns with the conclusion drawn from Question 2, where the confidence interval of the ATE doesn't include 0.

Question 4 (Amber)

In this question we are investigating the effect of eligibility for enrolling in a 401(k) plan on net financial assets in function of age, income and years of education, so we are looking for a heterogenous effect of eligibility. Again we (manually) applied 5-fold cross-fitting to obtain predictions for the nuisance parameters (outcome and the propensity scores) given all covariates mentioned in Question 2, to obtain weights $(A_i - \hat{E}(A_i|L_i))^2$ and pseudo-outcomes $\frac{Y_i - \hat{E}[Y_i|L_i]}{A_i - \hat{E}[A_i|L_i]}$ for each individual. Next, we fitted a `SuperLearner` model regressing these pseudo-outcomes on age, income and years of education to obtain individual CATE estimates, but also on each variable separately to visualize the effect in function of the corresponding variable, using these weights.



(a) (b)

(c) (d)

Figure 3: Individual CATE estimates (a) and effect estimates in function of age (b), income (c) and years of education (d).

The individual effect estimates are shown in the histogram in Figure 3a. We notice there are some extreme values among the effect sizes. The effect in function of age, income and years of education are visualized in Figure 3b, 3c and 3d respectively. Normally we would have to use cross-fitting for the last step (like we do in question 5) to obtain individual CATE estimates, as in ideal settings we would want to make predictions on new observations. For the purpose of this question however, considering the computational burden, we used the same data to train a model and make predictions in the last step. We see that the effect of eligibility on net financial assets increases with increasing age, income and years of education. However, we noticed that the effect sizes in function of income can take on some extreme values. This makes it harder to see an increasing effect, but this also has to do with the large scale of the income variable itself. Basicly a large increase in income will result in a large increase in the effect of eligibility.

Question 5 (Feng)

Algorithm for Question 5

To clearly illustrate the basic algorithm to estimate DR-learner in this context, we give the following procedures:

Step 1: split the data into 3 parts: like I, II, III.

Step 2: use part I to estimate nuisance functions, like $\hat{p}$ and $\hat{Y}^1$; use part II to predict and construct pseudo outcome $C_i = \hat{Y}_i^1 + \frac{A_i}{\hat{p}_i}(Y_i - \hat{Y}_i^1)$; swap the role of I and II.

Step 3: combine part I, II and use them together to train the model $E(C \mid \text{age, income, years of education})$; use part III to obtain the predictions $\tilde{Y}^1$.

Step 4: repeat Step 2 and 3 to conduct 3-folds cross-fitting for the estimation of $\tilde{Y}^1$.

Figure 4 shows the distribution of predictions $\tilde{Y}^1$ obtained via DR-learner, where we notice that DR-learner also has some extreme values.

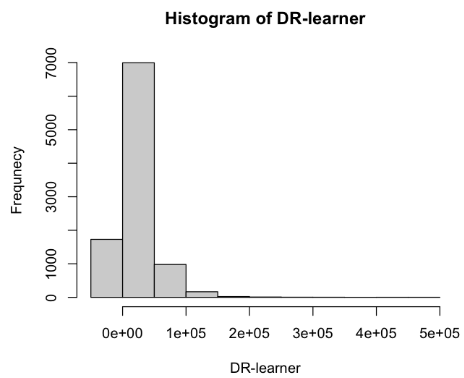


Figure 4: Predictions $\tilde{Y}^1$ based on DR-learner

To estimate this counterfactual prediction error $E\{(Y^1 - \tilde{Y}^1)^2\}$, we can treat $(Y^1 - \tilde{Y}^1)^2$ as a single counterfactual variable and then transfer the strategies of deriving the AIPW estimator of the ATE to estimate this prediction error. By this means, denoting $Z := (Y - \tilde{Y}^1)^2$, the estimator of $E\{(Y^1 - \tilde{Y}^1)^2\}$ is

$$\frac{1}{n}\sum_{i=1}^{n}\hat{E}(Z_i \mid A_i = 1, L_i) + \frac{1}{n}\sum_{i=1}^{n}\frac{A_i}{\hat{E}(A_i \mid L_i)}\{Y_i - \hat{E}(Z_i \mid A_i = 1, L_i)\}.$$

A 5-fold cross-fitting technique is used for computing the estimate above.

The estimate of the counterfactual prediction error is 3,045,283,867 with a 95% confidence interval [1,829,195,472; 4,261,372,263].